

BEHOLD!

THE PORTRAITRON 3000

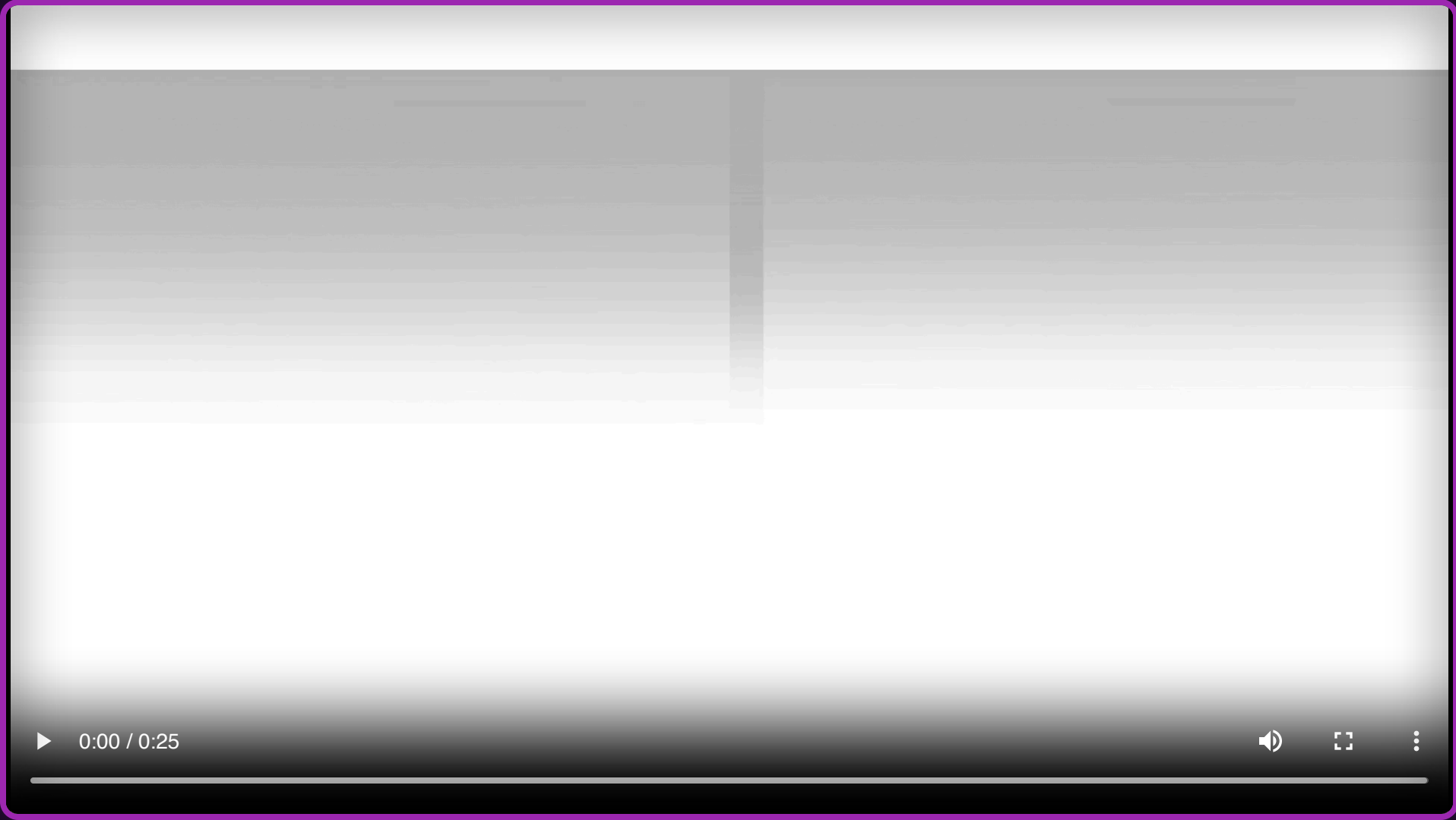
PROOF OF CONCEPT (POC) PHASE UPDATE

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IN MEMORIAM: THE FALLEN TOOLS

TRAGIC BACKSTORY



REMINDER: PROJECT FUNCTIONALITY & PLAN

WHAT WE PLAN TO DO

- **Capture high-definition picture** of the subject (3s delay)
- **Convert camera input** to vectors & stroke trajectories
- **Optimize strokes order** utilizing our sorting algorithm
- **Pick up & draw** strokes on page (Main POC Objective)
- **Stamp the page** with a commemorative stamp & hand over (Nice to have)

TRAGIC BACKSTORY

RADAR 3000
SCANNING...

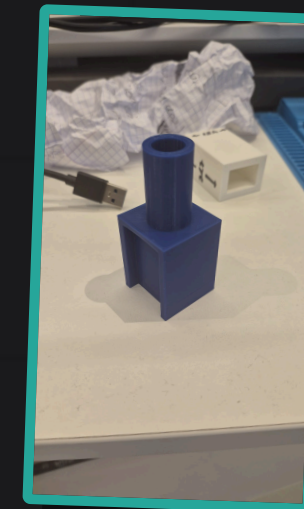
UPDATES: MOUNT DESIGN & APPROACH FORCE

1. PEN MOUNT EVOLUTION

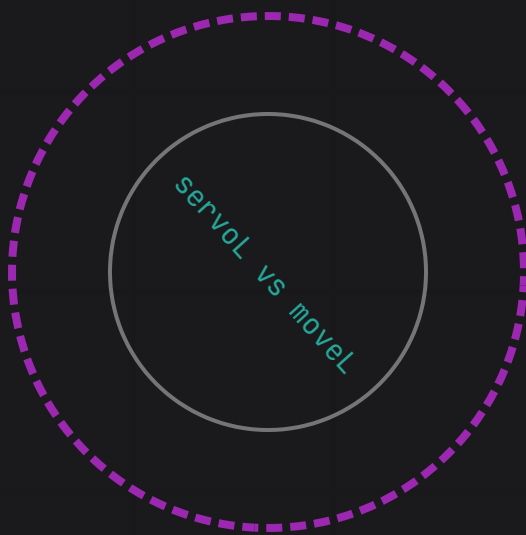
We successfully designed & printed the custom pen mount! However... **it doesn't fit the current drawing tool.**

2. MAINTAINING CONSTANT FORCE

Approaching the page: Originally done using **force mode**, but now done rigidly applying **manual control**. Our hypothesis is that force mode didn't work because of the velocity parameter configuration. We may try force mode again in the near future!



UPDATES: MOVEMENT & PAGE CONTROL



REVIEWED MOVEMENT MODES

To maintain absolute control while moving on the page surface, we thoroughly reviewed the core UR5e control behaviors:

moveL (Linear Position Control)

Position control receives higher priority than force mode control. Highly predictable but less compliant with minor paper surface warps.

servoL (Real-Time Control)

Allows you to inject positions dynamically in real-time. We discovered that force mode control works best when combined with real-time servo loops!

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CHALLENGES: THE INFAMOUS INK STAIN

THE LINGERING MARKER PROBLEM

While validating that the pen has reached the page (filtering out high force values caused by noisy sensor readings), the marker lingers at the starting coordinate for the contact validation period.

"Behold! The Unintentional Blot-Inator! It stains the paper before the drawing even begins!"

OUR SOLUTION PATH

We want to find a clean way to get rid of this initial dwell time blot.

- Change our drawing tool to a less bleed-heavy medium (e.g. ballpoint, pencil, or spring-loaded tip).
- We have the ability to calculate the touch plane using empirical numbers. Using this, we can reduce the error space of the force sensor to be 0.5 millimeters away from the drawing plane.

TRAGIC BACKSTORY

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PROOF OF CONCEPT GOALS REACHED

POC ACCOMPLISHMENTS

Approaching the page: Successfully approached the page until point of contact, without breaking the tip.

Semicircle Vector Execution: Successfully generated and drew an automated circle/semicircle trajectory, while maintaining a constant force on the page surface, despite wall angle.

Multiple stroke drawing: Successfully executed sequential strokes on the page, both linear and curved.



RADAR 3000
SCANNING...



TRAGIC BACKSTORY

FEASIBILITY & FUTURE PLANS

WHAT WE LEARNED

The application is **highly feasible!** The UR5e arm is precise enough to draw complex high-definition lines on paper.

However, noisy force feedback and physical drawing tool behavior (like initial bleed) require careful control loop optimization to ensure a clean visual finish.

DID OUR PLANS CHANGE?

NOPE!

We still fully intend to do everything we originally planned! All milestones, from SwiftSketch integration to full face vector drawing, remain on track.

What we don't know yet:

We haven't begun working on camera integration, page handing, and tool grabbing and dropping.

TRAGIC BACKSTORY



THE TRI-STATE AREA IS ALMOST OURS!

THANKS FOR LISTENING!

